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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2010 question paper for the guidance of teachers

5070 CHEMISTRY

5070/31 Paper 3 (Practical Test), maximum raw mark 40

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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1 **(a)** Titration

Accuracy 8 marks
For the two best titres give:
4 marks for a value within 0.2 cm^3 of supervisor
2 marks for a value within 0.3 cm^3 of supervisor
1 mark for a value within 0.4 cm^3 of supervisor

Concordance 3 marks
Give:
3 marks if all the ticked values are within 0.2 cm^3
2 marks if all the ticked values are within 0.3 cm^3
1 mark if all the ticked values are within 0.4 cm^3

Average 1 mark
Give 1 mark if the candidate calculates a correct average (error not greater than 0.05) of all his ticked values.

Assuming a 25 cm^3 pipette and a titre of 24.8 cm^3.

(b) concentration of hydrogen ions in **P** [2]

$= \frac{25 \times 0.1}{24.8}$ (1)

= 0.101 (1)

Answers should be correct to + or – 1 in the third significant figure.

(c) moles of hydrogen ions in 10000 dm^3 of contaminated water [1]

= 0.101 × 10000 (1)
= 1010

(d) mass of calcium carbonate needed to neutralise the acid [2]

= 1010/2 (1)

= 1010 × 100 (1)

= 50500 g

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2 **R** is aluminium **S** is potassium nitrate

Test	Notes
General points For ppt allow solid, suspension, powder For gases Name of gas requires test to be at least partially correct. Effervesces = bubbles = gas vigorously evolved (but not just gas evolved) Solutions Colourless not equivalent to clear, clear not equivalent to colourless	
Solution **R**	
Test 1 effervescence (1) pops with a lighted splint (1) hydrogen (1)	
Test 2 white ppt (1) soluble in excess (1) colourless solution (1)	
Test 3 white ppt (1) insoluble in excess (1)	
Test 4 **(a)** effervescence (1) pops with a lighted splint (1) hydrogen (1) **(b)** white ppt (1) soluble in excess (1) colourless solution (1)	
Test 5 **(a)** no reaction (1) **(b)** red/brown solid formed (1) blue colour fades (1) effervescence (1)	

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Test 6 **(a)** liquid turns green (1) **(b)** green ppt (1) insoluble in excess (1)		accept green-yellow or colourless black/dirty green ppt
Test 7 turns litmus blue (1) ammonia (1)		

[20]

R is aluminium/A*l* (ppt must dissolve in test 2 and ppt must not dissolve in test 3) (1)

R is acting as a reducing agent (any green in test **6(a)** or green/black in test **6(b)** (1)
S contains nitrate or NO_3^- (test 7 correct – allow alkaline gas, smell of ammonia) (1) [3]

Note: 26 marking points, maximum 23.